

Week-7 Day-2 Home assignment

Environment file :

MONGODB_URI=mongodb+srv://ninahire:Manchali2628@cluster0.bxtk4w9.mongodb.net/?appName=Cluster0

```
DB_NAME="Rag_testcases"
COLLECTION_NAME="testcases"
VECTOR_INDEX_NAME="vector_index_testcases"
BM25_INDEX_NAME="bm25_search"
USER_STORIES_COLLECTION_NAME="Userstories"
USER_STORIES_VECTOR_INDEX_NAME="vector_index_userstory"

# Mistral AI Configuration (for embeddings)
MISTRAL_API_KEY=pjXDTs7jiAEkd7rPTqhMq3XYXhTLLfcj
MISTRAL_EMBEDDING_MODEL="mistral-embed"
```

Get vector index ready in Mongo DB under Same DB with new collection added for Userstory ingestion

Search & Vector Search

Find a search index

VIEW COMPOUND QUERY EXAMPLE

You have reached the 3 search and vector index limit for your M0 free tier cluster. [Upgrade Cluster](#)

Database	Collection	Index Name	Status	Queryable	Type	Index Fields	Documents (Estimated)	Size	Recon Memo
Rag_testcases	Userstories	vector_index_userstories	READY View status details	✓	vectorSearch	"embedding", "jiraMetadata.projectKey", "status.name", "priority.name", "assignee.displayName", 3 more	0 (0%) indexed of 0	320B	320B

```
{
  "fields": [
    {
      "type": "vector",
      "path": "embedding",
      "numDimensions": 1024,
      "similarity": "cosine"
    },
    {
      "type": "filter",
      "path": "jiraMetadata.projectKey"
    }
  ]
}
```

```

{
  "type": "filter",
  "path": "status.name"
},
{
  "type": "filter",
  "path": "priority.name"
},
{
  "type": "filter",
  "path": "assignee.displayName"
},
{
  "type": "filter",
  "path": "components"
},
{
  "type": "filter",
  "path": "labels"
},
{
  "type": "filter",
  "path": "fixVersions"
}
]
}

```

At the terminal run npm install

```

ninahire@Ninas-Air rag-mongo-demo-ui % npm install
npm audit fix --force

Run `npm audit` for details.
npm warn allow-scripts 3 packages have install scripts not yet covered by allowScripts:
npm warn allow-scripts core-js@3.45.1 (install: (install scripts present))
npm warn allow-scripts core-js-pure@3.45.1 (install: (install scripts present))
npm warn allow-scripts fsevents@2.3.3 (install: (install scripts present))
npm warn allow-scripts Run `npm approve-scripts --allow-scripts-pending` to review, or `npm approve-scripts <pkg>` to allow.

up to date, audited 203 packages in 4s

31 packages are looking for funding
  run `npm fund` for details

14 vulnerabilities (5 moderate, 8 high, 1 critical)

To address issues that do not require attention, run:
  npm audit fix

Some issues need review, and may require choosing

```

Then run excel to userstories conversion script with this command
node src/scripts/data-conversion/excel-to-userstories.js

```
ninahire@Ninas-Air rag-mongo-demo-ui % node src/scripts/data-conversion/excel-to-userstories.js
✓ Valid user stories: 190
⚠ Filtered out: 0 empty rows

🚀 SUCCESS! Converted user stories from Excel to JSON
📁 Input: src/data/userstories.xlsx (sheet: "stories")
📁 Output: src/data/stories.json
📊 Total user stories: 190

📋 Sample converted user story:
Key: HC-257
Summary: Consultant Dashboard My Inpatients
Status: To Do
Priority: P1
Story Points: Not set
Project: Health Care

💡 Next steps:
1. Review the generated src/data/stories.json file
2. Run: node src/scripts/create-userstories-embeddings-batch.js
3. Create vector index in MongoDB Atlas for user stories search
ninahire@Ninas-Air rag-mongo-demo-ui %
```

Then run embedding script at terminal

node src/scripts/embeddings/[create-userstories-embeddings-batch-mistral.js](#)

```
ninahire@Ninas-Air rag-mongo-demo-ui % node src/scripts/embeddings/create-userstories-embeddings-batch-mistral.js
✓ [Batch 1/4] Success! Cost: $0.000813 | Tokens: 8128
✓ [Batch 3/4] Success! Cost: $0.000806 | Tokens: 8060
⏱ Waiting 1000ms before next batch group...
🚀 [Batch 4/4] Processing 40 user stories with Mistral AI...
✓ [Batch 4/4] Success! Cost: $0.000641 | Tokens: 6413
📊 Progress: 190/190 (100.0%) | Rate: 32.2/sec | ETA: 0s | Cost: $0.003068

🚀 MISTRAL AI BATCH PROCESSING COMPLETE!
📊 Final Statistics:
🕒 Total Time: 6s
⚡ Processing Rate: 29.4 user stories/second
📊 Total User Stories: 190
✓ Successfully Processed: 190
✗ Failed: 0
📈 Success Rate: 100.0%
💰 Total Cost: $0.003068
📊 Total Tokens: 30,678
📊 Average Cost per User Story: $0.00001615
📊 Average Tokens per User Story: 161
💡 Cost Savings: ~90% cheaper than OpenAI embeddings
ninahire@Ninas-Air rag-mongo-demo-ui %
```

Then check the mongo db ingestion for userstories

```
{
  "_id": ObjectId('6a85c0720e30f7870ef842ae'),
  "key": "HC-257",
  "summary": "Consultant Dashboard My Inpatients",
  "description": "As a user, I want to consultant dashboard my inpatients so that the re...",
  "status": Object(2),
  "priority": Object(2),
  "assignee": null,
  "reporter": null,
  "created": "2022-01-06T00:00:00.000Z",
  "updated": "2026-08-19T14:36:57.080Z",
  "components": Array(empty),
  "labels": Array(empty),
  "fixVersions": Array(empty),
  "storyPoints": null,
  "project": "Health Care",
  "epic": "Emergency Management",
  "acceptanceCriteria": "Given a user with valid access, when they open the dashboard, then rel...",
  "businessValue": "",
  "risk": "Critical",
  "dependencies": "",
  "notes": "",
  "sprint": null,
  "team": null,
  "issueLinks": Array(empty),
  "url": ""
}
```